means of monitoring loss of stored energy in hydraulic accumulators. These alarms are to be audible and visual and shall be situated on the central operating console at the navigation bridge; or

.2 an independent hydraulic system for each door with each power source consisting of a motor and pump capab le of opening and closing the door. In addition, there shall be a hydraulic accumulator of sufficient capacity to operate the door at least three times, i.e. closed-open-closed, against an adverse list of 15 o. This operating cycle shall be capable of being carried out when the accumulator is at the pump cut-in pressure. The fluid used shall be chosen considering the temperatures liable to be encountered by the installation during its service. A low gas pressure group alarm or other

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
shall be chosen considering the temperatures liable to be encountered by the installation during its service. A low gas pressure group alarm or other effective means of monitoring loss of stored energy in hydraulic accumulators shall be provided at the central operating console on the navigation bridge. Loss of stored energy indication at each local operating position shall also be provided; or

.3 an independent electrical system and motor for each door with each power source consisting of a motor capable of opening and closing the door. The power source shall be capable of being automatically supplied by the transitional source of emergency electrical power as required by
RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-
39 -

regulation 42.4.2 in the event of failu re of either the main or emergency

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-
39 -

regulation 42.4.2 in the event of failu re of either the main or emergency source of electrical power and with suffi cient capacity to operate the door at least three times, i.e. closed-open-closed, against an adverse list of 15o.

For the systems specified in paragraphs 7.3.1, 7.3.2 and 7.3.3, provision should be made as follows: Power systems for power-operated watertight sliding doors shall be separate from any other power system. A single failure in the electric or hydraulic power-operated systems excluding the hydraulic actuator shall not prevent the hand operation of any door.

7.4 Control handles shall be provided at each side of the bulkhead at a minimum height of 1.6 m above the floor and shall be so arranged as to en able persons passing through the doorway to hold both handles in th e open position without being able to set

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
height of 1.6 m above the floor and shall be so arranged as to en able persons passing through the doorway to hold both handles in th e open position without being able to set the power closing mechanism in operation acci dentally. The direction of movement of the handles in opening and clos ing the door shall be in the direction of door movement and shall be clearly indicated.

7.5 As far as practicable, electrical equi pment and components for watertight doors shall be situated above the bulkhead deck and outside hazardous areas and spaces.

7.6 The enclosures of electrical components necessarily situated below the bulkhead deck shall provide suitable protection against the ingress of water.

7.7 Electric power, control, indication and alarm circuits shall be protected against fault in such a way that a failure in one door circuit will not cause a failure in any other door circuit. Short circuits or other faults in the alarm or indicator circuits of a door shall

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
fault in such a way that a failure in one door circuit will not cause a failure in any other door circuit. Short circuits or other faults in the alarm or indicator circuits of a door shall not result in a loss of power operation of that door. Arrange ments shall be such that leakage of water into the el ectrical equipment located below the bulkhead deck will not cause the door to open.

7.8 A single electrical failure in the power operating or control system of a power-operated sliding watertight door shal l not result in a closed door opening. Availability of the power supply should be continuously monitored at a point in the electrical circuit as near as practicable to each of the motors required by paragraph 7.3. Loss of any such power supply should activate an audible and visual alarm at the central operating console at the navigation bridge.

8.1 The central operating consol e at the navigation bridge shall have a master mode

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
operating console at the navigation bridge.

8.1 The central operating consol e at the navigation bridge shall have a master mode switch with two modes of cont rol: a local control mode which shall allow any door to be locally opened and locally closed after use without automatic closure, and a doors closed mode which shall automatically close any door that is open. The doors closed mode shall automatically close any door that is open. The doors closed mode shall permit doors to be opened locally and shall automatically re-close the doors upon release of the local control mechanism. The mas ter mode switch shall normally be in the local control mode. The doors closed mode shall only be used in an emergency or for testing purposes. Special consideration shall be given to the reliability of the master mode switch.

8.2 The central operating console at the na vigation bridge shall be provided with a

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
for testing purposes. Special consideration shall be given to the reliability of the master mode switch.

8.2 The central operating console at the na vigation bridge shall be provided with a diagram showing the location of each door, with visual indicators to show whether each door is open or closed. A red light shall i ndicate a door is fully open and a green light shall indicate a door is fully cl osed. When the door is closed remotely the red light shall indicate the intermediate position by flashing. The indicating circuit shall be independent of the control circuit for each door.
RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-40 -

8.3 It shall not be possible to remo tely open any door from the central operating console.

9.1 If the Administration is satisfied that such doors are e ssential, watertight doors of

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
8.3 It shall not be possible to remo tely open any door from the central operating console.

9.1 If the Administration is satisfied that such doors are e ssential, watertight doors of satisfactory construction may be fitted in watertight bulkheads dividing cargo between deck spaces. Such doors may be hinged, rolling or sliding doors but shall not be remotely controlled. They shall be fitted at the highest level and as far from the shell plating as practicable, but in no case shall the outboard vertical edges be situated at a distance from the shell plating which is less than one fifth of the breadth of the ship, as defined in regulation 2, such distance being measured at ri ght angles to the centreline at the level of the deepest subdivision draught.

9.2 Should any such doors be accessible during the voyage, they shall be fitted with a device which prevents unauthorized opening. When it is proposed to fit such doors, the

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
9.2 Should any such doors be accessible during the voyage, they shall be fitted with a device which prevents unauthorized opening. When it is proposed to fit such doors, the number and arrangements shall receive the special consideration of the Administration.

10 Portable plates on bulkheads shall not be permitted except in machinery spaces. The Administration may permit not more than one power-operated sliding watertight door in each watertight bulkhead larger than those specified in paragraph 7.1.2 to be substituted for these portable plates, provide d these doors are intended to remain closed during navigation except in case of urgent necessity at the discretion of the master. These doors need not meet the requirements of paragraph 7.1.4 regarding complete closure by

hand-operated gear in 90 s.

11.1 Where trunkways or tunnels for access from crew accommodation to the stokehold, for piping, or for any other purpos e are carried through watertight bulkheads,

---CHUNK---

SOURCE: MSC.216%2882%29.pdf
hand-operated gear in 90 s.

11.1 Where trunkways or tunnels for access from crew accommodation to the stokehold, for piping, or for any other purpos e are carried through watertight bulkheads, they shall be watertight and in accordance with the requirements of regulation 16-1. The access to at least one end of each such tunnel or trunkway, if used as a passage at sea, shall be through a trunk extending watertight to a height sufficient to permit access above the bulkhead deck. The access to the other end of the trunkway or tunnel may be through a watertight door of the type required by its location in the ship. Such trunkways or tunnels shall not extend through the first subdivision bulkhead abaft the collision bulkhead.

11.2 Where it is proposed to fit tunnels pi ercing watertight bu lkheads, these shall receive the special consideration of the Administration.

11.3 Where trunkways in connection with refr igerated cargo and ventilation or forced

---CHUNK---